

Advanced ecological models

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The real world is not easy (1)

Although all previous models can be very helpful, **in real life things are usually more complicated**:

- Sometimes data do not come from independent data because the **sampling is not random** but **preferential**. Like fishermen who go fishing where they think there is fish, so clever option (Pennino et al., 2019).
- Sometimes there is **no available information in those places we want to make predictions**: **misalignment** (Barber et al., 2016).
- The above presented approach does not take into account that **spatial data are often multivariate**. Consequently, a **multivariate geostatistical approach** should be used (Barber et al., 2019; Barber et al., 2021).
- In other situations, the **spatial dependence structure is NOT the same** throughout the time considered and different **spatio-temporal structures** can be considered (Paradinas et al., 2017).
- Another issue appears when there are **more zeros than the proposed distribution could reasonably explain**, and so, **zero inflated** or **hurdle** models are needed (Paradinas et al., 2017).

The real world is not easy (2)

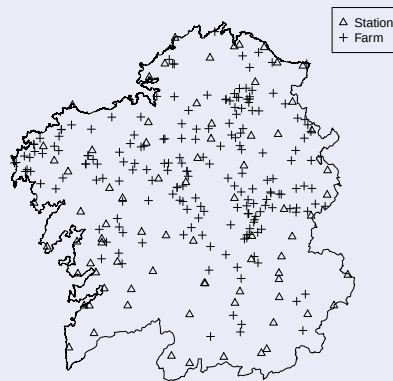
- When geographical elements or physical **barriers** are present, the **spatial dependence structure could NOT be the same** throughout the domain: **non-stationarity** (Martínez-Minaya et al., 2019; Cendoya et al., 2022).
- In some occasions, we can have **information from varios sources** and want to **join them in one unique modelling** (Figueira et al., 2024a; Figueira et al., 2025).
- An important situation is how to deal with **only-presence** data (Martino et al., 2021, Fernández et al., 2025).
- See Martínez-Minaya et al. (2018) for a **review of some of these issues**.
- See also Figueira et al. (2024b) for a **Shiny app** that allows to use most of the procedures here mentioned in an easy way.

1 Misalingment

Dealing with misalignment

- It appears when **measurement values of the covariates are not known** at the observed locations nor at those locations where we are going to make predictions.

Example of misalignment: the 67 official weather stations in Galicia do not coincide with the farms where data were observed



Dealing with misalignment (2)

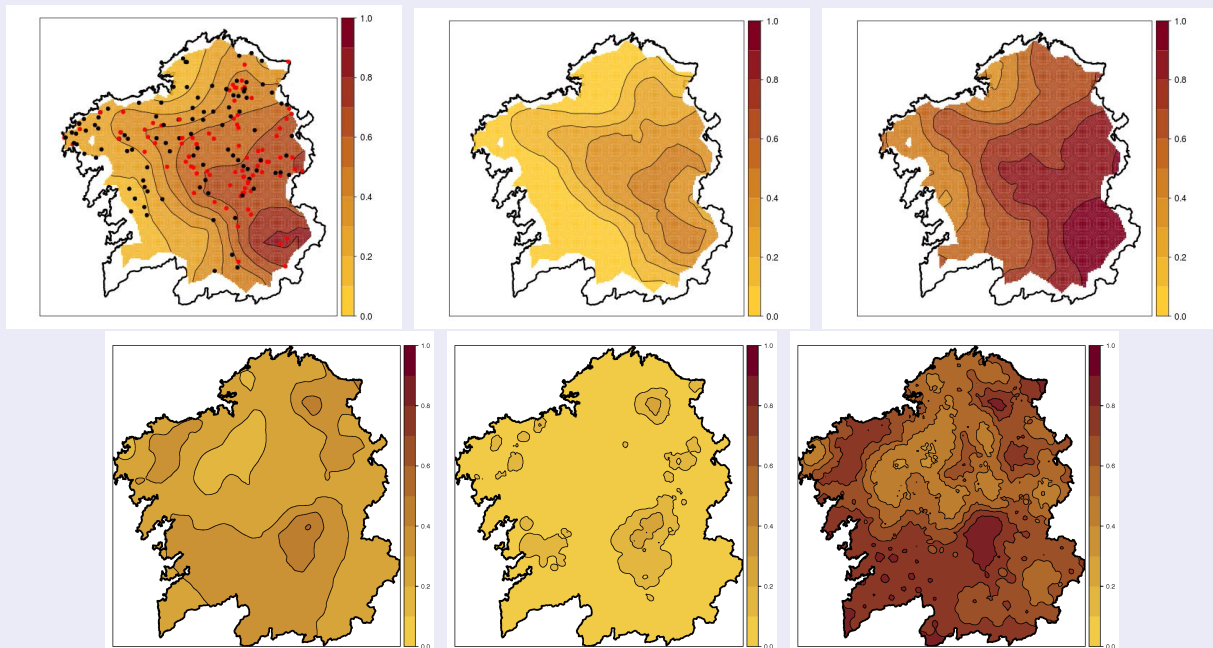
- A **spatial geostatistical model is specified for the covariate**, and it is **estimated jointly with the species distribution models in a Bayesian context**.
- The joint model is specified as follows

$$\begin{aligned}y_i &\sim F(\mu_i) \\g(\mu_i) &= \beta_0 + \beta_1\phi_i + w_i \\w &\sim N(0, Q(\kappa, \tau)) \\x_i &\stackrel{iid}{\sim} N(\phi_i, \sigma_x^2) \\\phi &\sim N(0, Q(\gamma, \delta))\end{aligned}$$

- ▶ μ_i is the mean of the observed variable of interest at site s_i ;
- ▶ x_i is the **covariate of interest whose spatial distribution is specified** through its mean (a realization of the Matérn Gaussian process ϕ depending on the parameters γ and δ), and variance σ_x^2 , which is introduced to express any possible measurement error; and
- ▶ w is the spatial process for the response.

Analysing the Fasciolosis in Galicia (Barber et al., 2016)

Posterior mean of the probability of occurrence (left) and the first (center) and third (right) quantiles. Red points are presences and black absences. Above without taking into account misalignment, below taking it into account.



2 Coregionalised approach for multivariate SDMs

Coregionalised approach for multivariate SDMs

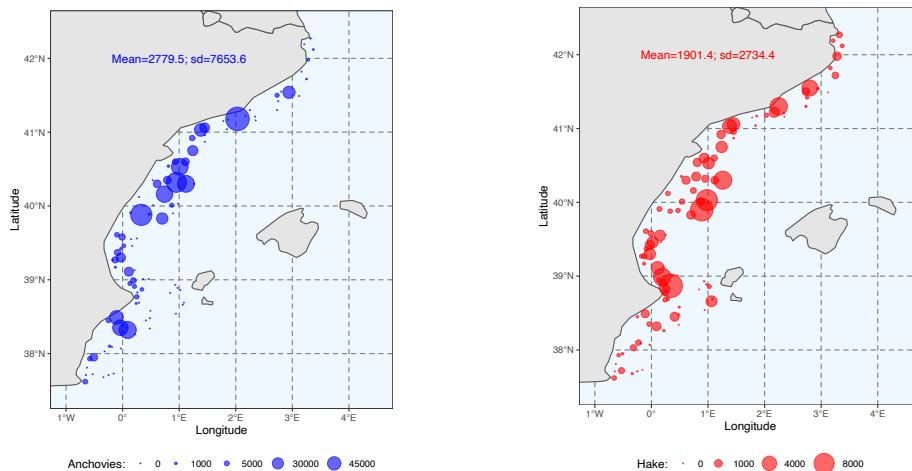
- SDMs are usually analyzed as univariate problems, not taking into account that **spatial data are often multivariate**.
- In **order to explain** the possible presence/absence or the abundance of a species, we must include in the modelling the **biotic relationships** with other species:
 - ▶ **competition**,
 - ▶ **predator-prey**,
 - ▶ **parasitism**, or
 - ▶ **mutualism**.
- **Despite its enormous relevance**, these factors are **usually ignored** in most classical SDM-based studies.
- Better than include these relationships via covariates, a great idea here is to use **coregionalised models**.

Coregionalised approach for multivariate SDMs (2)

- Information about the distribution of marine species:
 - ▶ **fishery-independent data** (**scientific surveys at sea**) and
 - ▶ **fishery-dependent data** (collection and sampling by observers in **commercial vessels**), preferential sampling.
- **Scientific survey data are considered to be of higher quality** (sampling designed to ensure unbiased estimations).
- But surveys may produce biased estimates:
 - ▶ for **species** with preferential habitats that are in strata only **partially included** in the survey sampling design,
 - ▶ or just for **catchability issues**.
- Again a good idea is to use **multivariate geostatistical coregionalised models**.
- **Species well represented** in the sampling could be **used to infer** the spatial behaviour of those **correlated but poorly represented**.

Describing interaction between anchovies and hake (Barber et al., 2021)

Application in fishery Ecology: a **prey-predator example** between anchovies and hake, one of its predator species, both collected during trawling surveys in the Mediterranean sea.



Data from EU-funded survey project MEDiterranean Trawl Survey (MEDITS): **map** shows the **abundance at each sampling location** of anchovies (**left**) and hake (**right**).

Coregionalised model for anchovies and hake

Coregionalised model that tries to reflect the predator-prey relationship between anchovies and hake species

$$\begin{aligned}\log(\text{Hake}) &= \beta_{10} + \beta_{11}\text{Bathymetry} + \mathbf{W}_1 + \epsilon_1 \\ \log(\text{Anchovy}) &= \beta_{20} + \beta_{21}\text{Bathymetry} + \alpha\mathbf{W}_1 + \mathbf{W}_2 + \epsilon_2\end{aligned}$$

where

- α represents the association between both species;
- $\mathbf{W}_1$ and $\mathbf{W}_2$ are the spatially structured effects;
- **linear relationship between species and bathymetry** (non-linear could also have been fitted by incorporating second-order random walk latent models);
- **non-informative Gaussian priors** for all fixed effects; and **PC-priors** to describe prior knowledge of hyperparameters of the spatial terms;
- no interacting univariate models $\alpha = 0$ were also analysed.
- **implemented in INLA** by **sharing the spatial term** in both likelihoods.

Coregionalised model for anchovies and hake: results

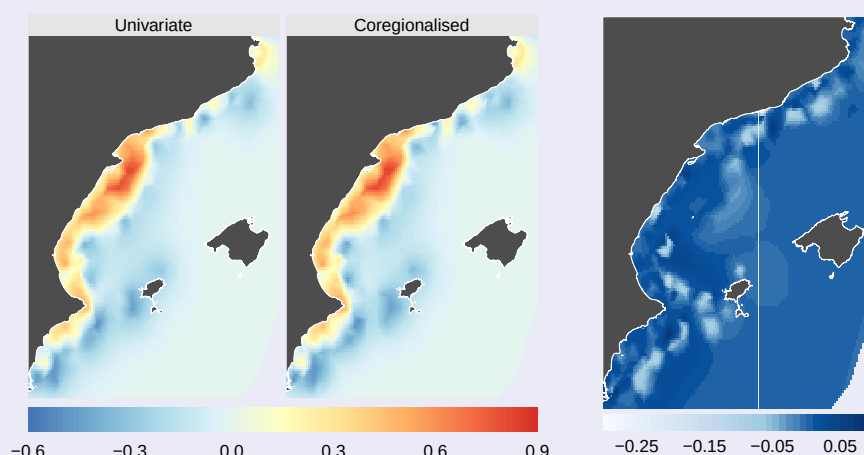
Coregionalised model that tries to reflect the predator-prey relationship between anchovies and hake species

	Mean	sd	$q_{0,025}$	$q_{0,5}$	$q_{0,975}$
β_{10}	6.5258	0.2759	5.9600	6.5343	7.0434
β_{20}	4.4426	0.3934	3.6091	4.4609	5.1692
β_{11}	-0.0038	0.0007	-0.0052	-0.0038	-0.0024
β_{21}	-0.0021	0.0008	-0.0036	-0.0021	-0.0006
ϕ_1	0.161	0.028	0.113	0.158	0.223
σ_1	2.148	0.152	1.864	2.143	2.463
ϕ_2	0.575	0.154	0.350	0.548	0.948
σ_2	1.544	0.194	1.212	1.527	1.971
α	0.143	0.073	-0.003	0.144	0.285

Association between hake and anchovy is positive: geographical patterns of abundance of both species are related.

Coregionalised model for anchovies and hake: results (2)

Maps of posterior mean of spatial effect in univariate model (left), coregionalised model (middle), and difference of two spatial effects (right) for anchovies



Although differences are not remarkable, they still show that coregionalised model provides different (better?) results than the univariate model.

Coregionalised model for anchovies and hake: results (3)

Predictive capacity of Univariate and coregionalised models obtained by cross-validation of real 2016 data with the previous adjusted model.

Model	Measure	Value
Univariate	MAE	4212.35
Coregionalisation	MAE	3842.35
Univariate	RMSE	8422.54
Coregionalisation	RMSE	7697.88

Coregionalised model shows lower values than the univariate model, both in terms of MAE and RMSE, indicating a better predictive capacity.

3 | Spatio-temporal structures

Spatio-temporal structures for SDMs

- In SDMs, most studies have been repeated periodically for long periods of time and so, there is interest in knowing whether the spatial evolution of the system under study varies not only in space but also in time.
- Temporal correlation depends on the same principle as spatial correlation: temporally close observations tend to be more related than temporally distant ones.
- Consequently, model fitting and predictions improve when a temporal term is added.
- However, temporal and spatial scales are different and the spatio-temporal analysis is more complicated than the simple addition of an extra dimension to the continuous spatial domain.

Spatio-temporal structures

Assuming a geostatistical spatial term ($\mathbf{w}$):

$$\mathbf{w} \sim N(0, \mathbf{Q}(\kappa, \tau))$$

There are different U_{st} structures to infer the fundamental temporal behaviour of the process. Among them and following Paradinas et al. (2017):

- Opportunistic spatial distribution

$$U_{st} = w_{st} \quad (1)$$

- $\simeq$ Persistent spatial pattern with uncorrelated intensities

$$\begin{aligned} U_{st} &= w_{st} + v_t \\ v_t &\sim N(0, \sigma_t) \end{aligned} \quad (3)$$

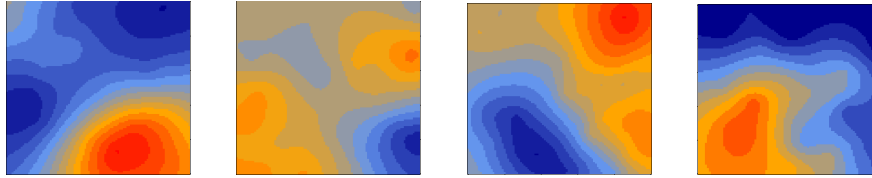
- Persistent spatial pattern with a temporal trend

$$U_{st} = w_{st} + f_t(t) \quad (2)$$

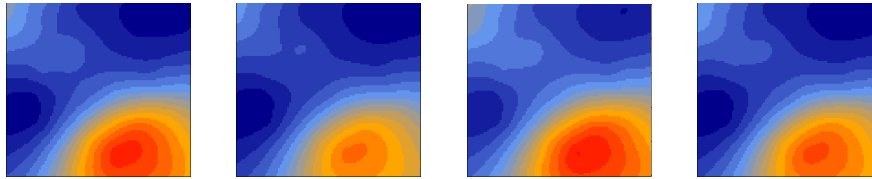
- Correlation among neighbouring years

$$\begin{aligned} U_{st} &= r_{st} + w_{st} \\ r_{st} &\sim N \left(\sum_k \rho_k r_{st-k}, \tau_r^{-1} \right) \end{aligned} \quad (4)$$

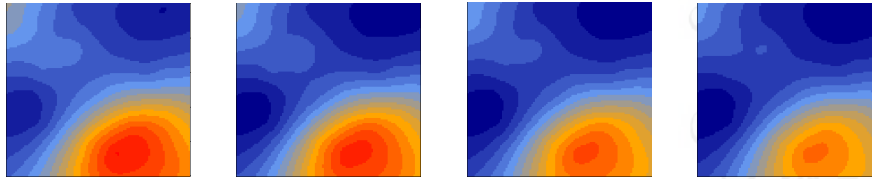
a) Opportunistic distribution



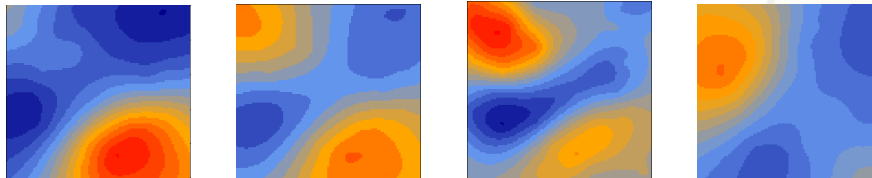
b) Persistent distribution with random changes



c) Persistent distribution with trend



d) Progressive spatio-temporal distribution



4 | Dealing with excess of zeroes

How to deal with excess of zeroes

- A source of overdispersion caused by a **disagreement between the data and the distribution assumed**: there are **more zeros than the proposed distribution could reasonably explain**.
- Different options:
 - ▶ **Zero-inflated models**: finite mixture of a degenerate distribution with all its mass at zero with a discrete distribution with support in $\mathbb{Z}^+ \cup \{0\}$.
 - ▶ **Hurdle models for count data**: a finite mixture of a degenerate distribution with all its mass at zero and a zero truncated discrete distribution. Unlike the previous, all observed zeros come from the zero-degenerate distribution.
 - ▶ **Hurdle models for semi-continuous processes** (rain, plant coverage, chemical concentrations, etc.) measured in the $[0, \infty)$ interval having high proportions of zero values are modeled as two independent sub-processes: one determines whether the response is zero, and the other determines the intensity when the response is non-zero using a continuous well known distribution (like Gamma).

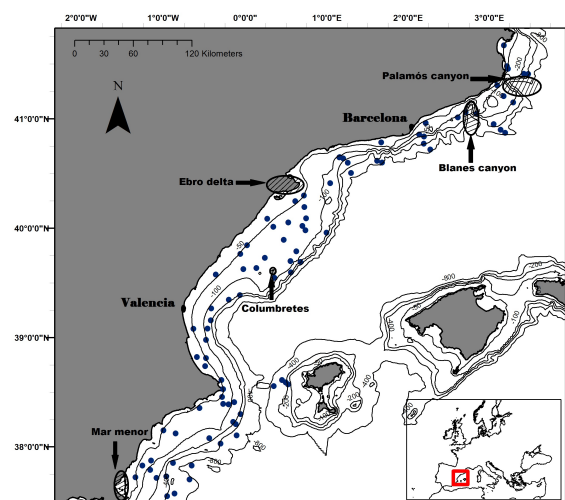
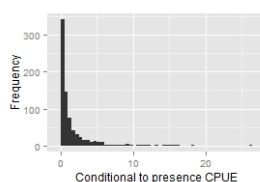
Analysis of hake recruitment (Paradinas et al., 2017)

Data:

- 1048 observations
- from 2000 to 2012
- Occurrence

Presence	Absence
758	290

- Excess of zeroes
- Conditional-to-presence Abundance



Bathymetry is the main known driving factor of hake juveniles. Preference to **80-250 meters** according to literature.

Two-part hurdle model for the abundance (continuous) of hake juveniles that allows different spatio-temporal structures in both distributions via shared components:

- Occurrence $\rightarrow Y_{st} \sim \text{Ber}(\pi_{st})$
 $\text{logit}(\pi_{st}) = \beta_o + f(\text{depth}) + u_{st}$
- Abundance $\rightarrow Z_{st} \sim \mathcal{Ga}(a_{st}, b_{st})$
 $\log(\mu_{st}) = \beta_a + \theta_f f(\text{depth}) + \theta_u u_{st}$
 - ▶ u_{st} corresponding a spatio-temporal structure
 - ▶ θ_f and θ_u are scaling parameters that link the shared component in the bathymetry and in the spatial effect
 - ▶ The latent Gaussian field for the bathymetry is random walk of order 2

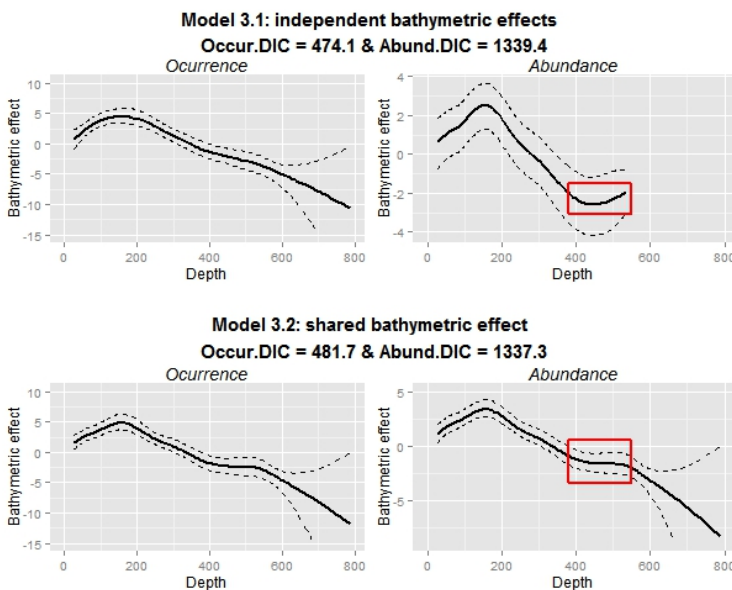
- Every combination of spatio-temporal structures and shared components were compared based on the deviance information criterion (DIC). Here's a summary:

Model	Structure	DIC	
		Occur	Abund
Model 0.1	I(b) + I(w) + I(iid.t)	466.8	1424.6
Model 0.2	I(b) + I(w) + I(trend.t)	475.9	1428.2
Model 1.1	I(b) + w_t	554.9	1487.6
Model 2.1	I(b) + S(w) + I(iid.t)	513.1	1432.8
Model 2.2	S(b) + I(w) + I(iid.t)	479.3	1425.4
Model 3.1	I(b) + I(w*t)	474.1	1339.4
Model 3.2	S(b) + I(w*t)	481.7	1337.3
Model 3.3	I(b) + S(w*t)	493.4	1573.7
Model 3.4	S(b) + S(w*t)	494.7	1573.7

b = bathymetry, w = spatial effect, t=temporal effect, w*t = spatio temporally structured effect. S() = shared components, I() = independent components.

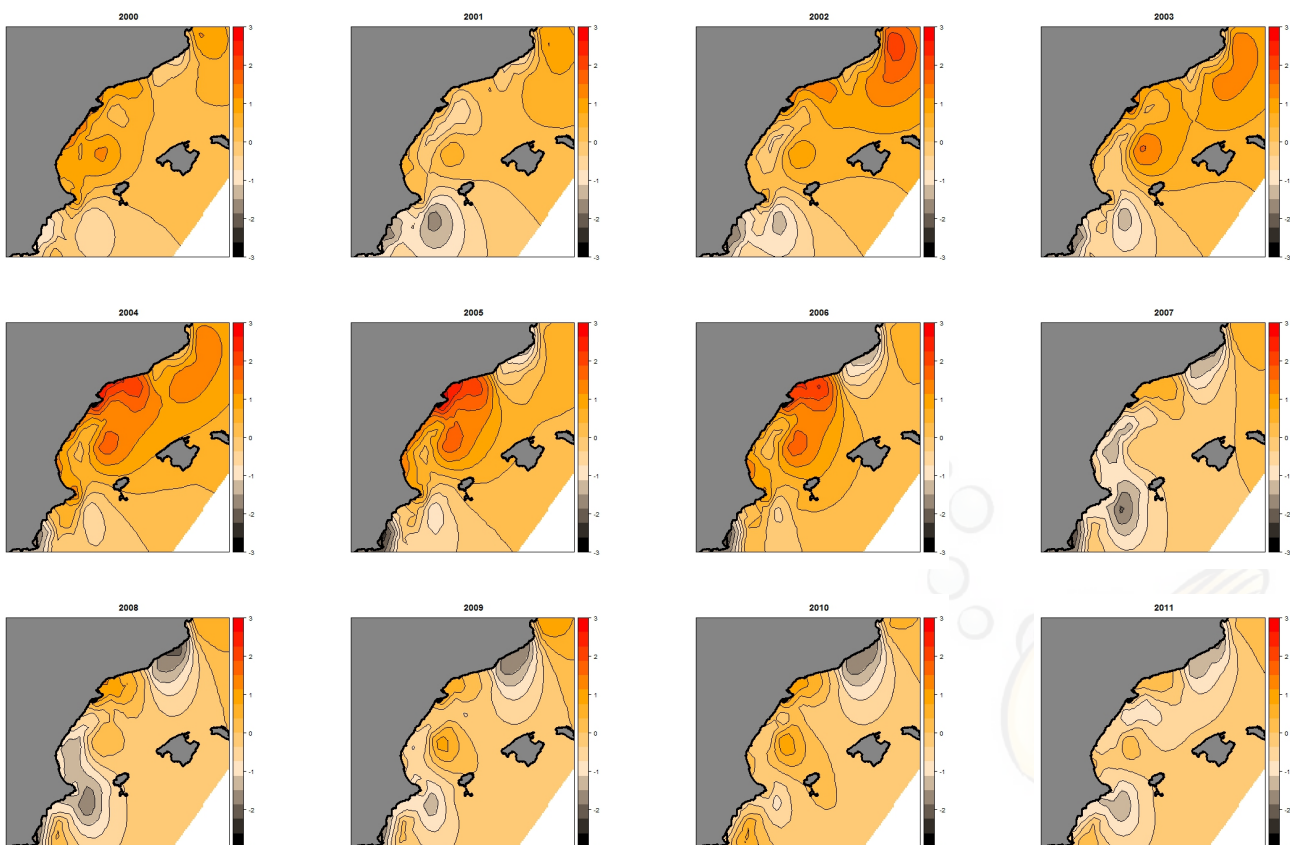
Results: shared bathymetric effect

- Finally $S(b) + I(w*t)$ was selected



- Model 3.1 slightly overfit the data (**box**). Still abundance DIC better in Model 3.2
- Model 3.2 fits a more natural bathymetric effect
- Model 3.2 allow the model predict deeper

Results: posterior distribution of the spatial effect for all years in the study

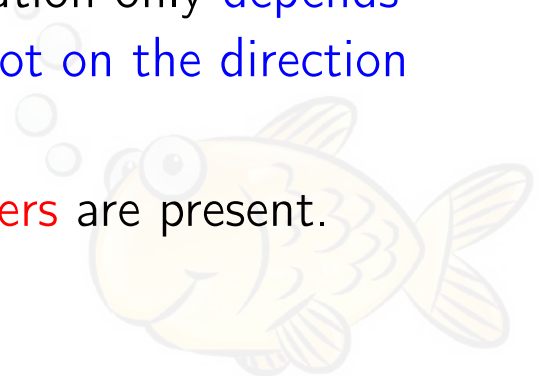


5 | The effect of barriers



Dealing with barriers in SDMs

- When geographical elements or physical **barriers** are present, the **spatial dependence structure could NOT be the same** throughout all the domain: **non-stationarity**.
- SDMs usually assume stationarity in the spatial random effect, that is, the spatial autocorrelation only **depends on the distance between locations, not on the direction nor the coordinates**.
- Can be **incorrect** when **physical barriers** are present.



Dealing with non-stationarity

- Bakka et al. (2019) proposed a way to construct a GMRF locally (with one governing equation for the normal area, and another for the barrier area) as the solution to the system of SPDEs:

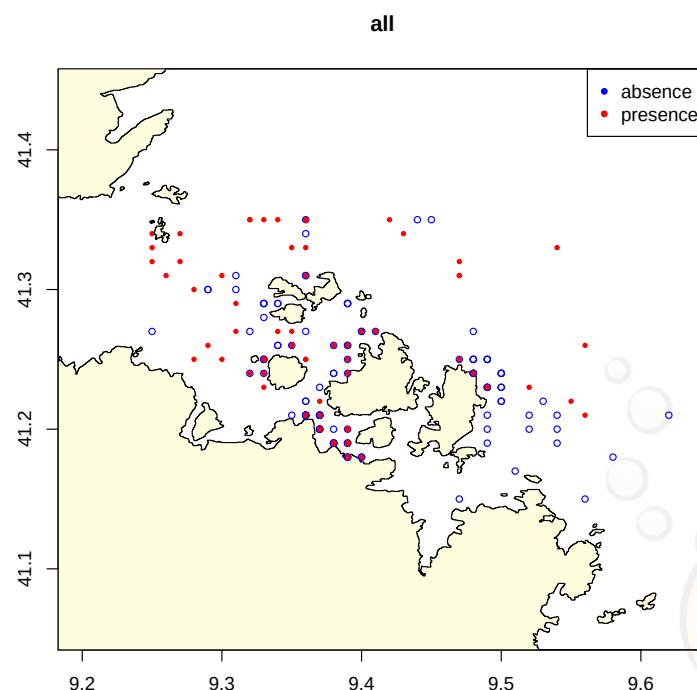
$$\begin{aligned}u(s) - \nabla \cdot \frac{r^2}{8} \nabla u(s) &= r \sqrt{\frac{\pi}{2}} \sigma_u W(s), \text{ for } s \in \Omega_n, \\u(s) - \nabla \cdot \frac{r_b^2}{8} \nabla u(s) &= r_b \sqrt{\frac{\pi}{2}} \sigma_u W(s), \text{ for } s \in \Omega_b\end{aligned}$$

where

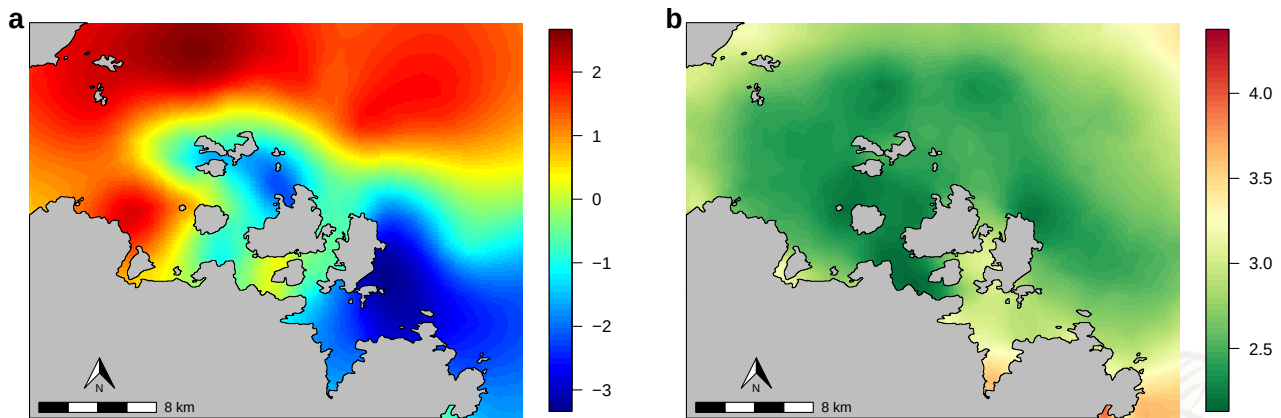
- ▶ Ω_n is the normal area; Ω_b is the barrier area; and
- ▶ the range r_b in the barrier area is fixed close to zero.
- Martínez-Minaya et al. (2019) and Panunzi et al. (2024) have analyzed fisheries barriers with this approach.

Presence of dolphins in the Madalena

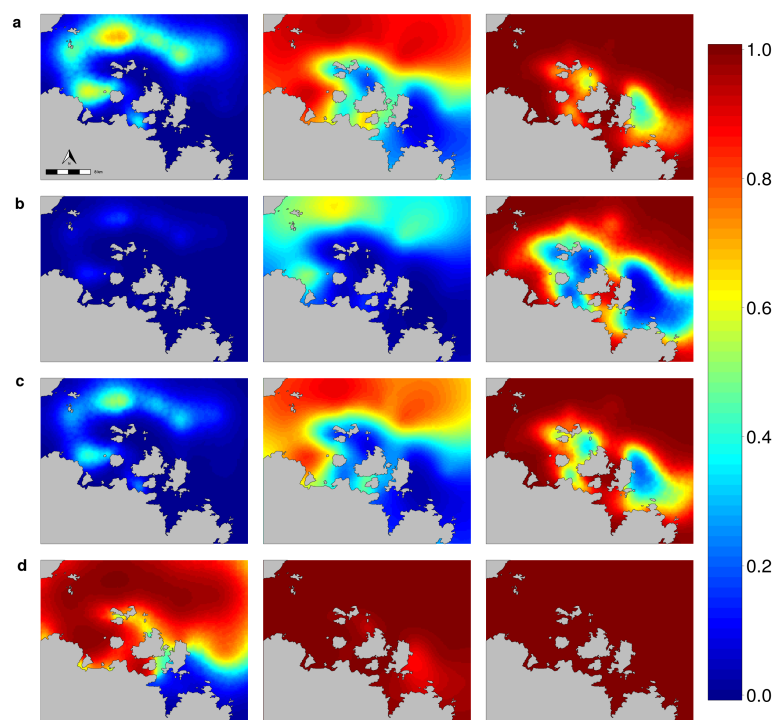
- Martínez-Minaya et al. (2019) have used this approach to analyse the presence of bottlenose dolphins in the North of Sardinia.



- Mean and standard deviation for posterior distribution of the **spatial effect w** .



- **Posterior predictive distribution of the probability of presence:** 95 % credible intervals (First and third panel respectively) and the median (central panel) for the different seasons. a: autumn, b: summer, c: spring and d: winter.



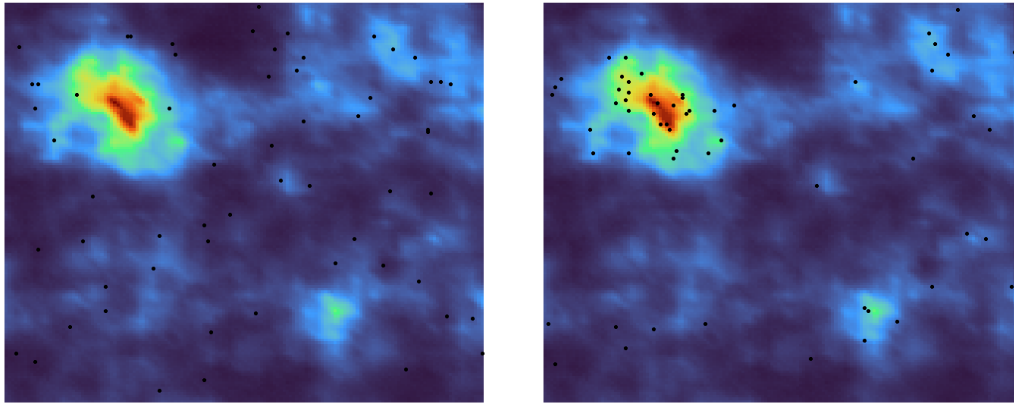
6 | Combining different sources of data for SDMs



Combining information in Environmental sciences.

- **Information** can be provided by means of **different sources** in order to explain the same phenomenon. For example:
 - ▶ **Dependent data**: the sampling is preferential, influenced by the process under study. Usually in on-board observation, citizen data.
 - ▶ **Independent data**: it's a random sampling (information not shared with the geostatistical process). Usually in scientific surveys.
- **Combining these diverse datasets** has become an essential tool for **managing the increasing complexity and volume of ecological data**.
- As data complexity and volume grow, **computational demands of previous proposals for data integration are also higher**, creating significant challenges for practical implementation.
- Combining information in complex **Spatio-temporal models** is a clear **example of this resulting computational burden**.

Example of two different sources of information



- Left side represents independent data collected from **scientific surveys** with a completely random sample pattern,
- while the right hand side represents dependent data coming from **opportunistic** or **preferential** sampling (e.g., whale watching or fishery commercial vessels),
- in which observers tend to look for a specific species or to fish in those areas where they expect to find it.

How to combine information

- Different **ways for combining sources of information**:
 - ▶ **Data pooling**, aggregating data without explicitly accounting for their diversity;
 - ▶ **Ensemble modelling**, where multiple diverse models are created to predict a single outcome, either by applying various individual models to the same dataset or by using a single modelling setup across different datasets<,
 - ▶ **Integrated models**, combining data by modelling various datasets simultaneously.

Integrated models.

- **Integrated models** allow different **sources of information to be combined by sharing components**:
 - ▶ used to **combine samples with different structures**, such as completely random samples, stratified random samples or preferential samples; or
 - ▶ can also be used to **combine information from different types of data** on variables that share components in the latent structure.
- **Examples**:
 - ▶ combining the number of catches of a species with other information on the abundance of the same species;
 - ▶ combining information on the presence/absence of a toxin along a river with other measures of its concentration at different locations
- By **combining the two sources of information in a joint model**, we can **analyse both variables simultaneously** and to **use common elements of the latent field** for a **more accurate and robust estimation**.

Integrated models (2).

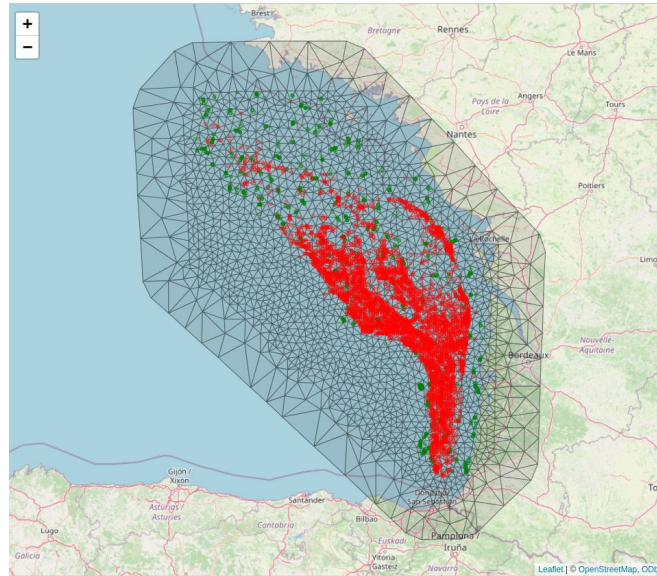
- In the previous examples, we have **two different likelihoods** whose linear predictors **share a spatial component** up to a parameter α :

$$\begin{aligned}y_{1i} \mid \eta_{1i}, \boldsymbol{\theta}_1 &\sim \ell_1(y_{1i} \mid \eta_{1i}, \boldsymbol{\theta}_1), \\y_{2j} \mid \eta_{2j}, \boldsymbol{\theta}_2 &\sim \ell_2(y_{2j} \mid \eta_{2j}, \boldsymbol{\theta}_2), \\g_1(\mu_{1i}) = \eta_{1i} &= \beta_{10} + \mathbf{A}_{1i}\boldsymbol{\beta}_1 + \sum_{j=1}^J f_{1j}(z_{1ij}) + u(s_i), \\g_2(\mu_{2j}) = \eta_{2j} &= \beta_{20} + \mathbf{A}_{2j}\boldsymbol{\beta}_2 + \sum_{j=1}^J f_{2j}(z_{2ij}) + \alpha \cdot u(s_j),\end{aligned}$$

where ℓ_1 and ℓ_2 are the likelihood functions for y_1 and y_2 , and $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ are the set of hyperparameters associated with each likelihood.

- Note the large **flexibility to analyze multiple sources of information** together.
- Resulting spatial models can be analysed with **INLA-SPDE**:
 - ▶ Inference on the parameters;
 - ▶ Model selection; and
 - ▶ Prediction on new locations (map of the quantity of interest all over the space).

Three data sets of info for the hake in the bay of Biscay



Case study of the distribution of hake in the Bay of Biscay during the years 2003 to 2021.

Three data sets of info for the hake in the bay of Biscay

- Information about the **distribution of hake** in the Bay of Biscay during the years 2003 to 2021 comes **via three datasets**:
 - ▶ the **scientific EVHOE trawl survey** which collected **discrete abundance data**;
 - ▶ one **commercial fishing fleet**, sampled by **on-board observers and targeting hake**, collected **continuous biomass data**; and
 - ▶ another **commercial fishing fleet**, sampled also by **on-board observers but not targeting hake**, recorded **presence-absence data**.
- Note that the **first commercial fleet targeting hake** carried out a **preferential exploration** of the sea in order to maximise the hake biomass catch.

Combining the different sources of information (3)

The **joint model** that takes into account all this information:

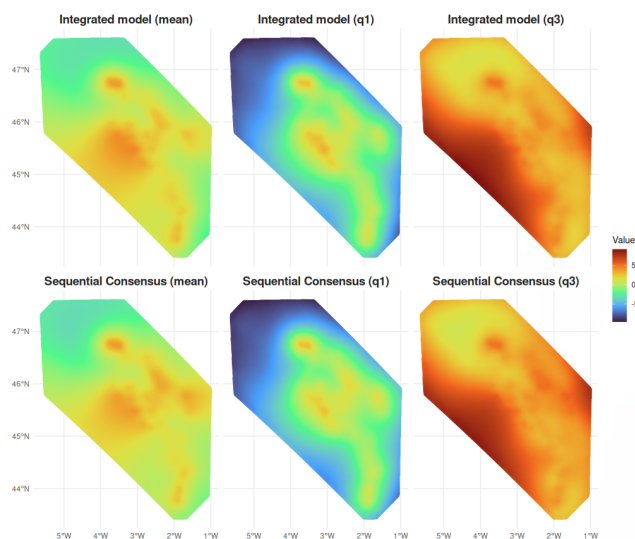
$$\begin{aligned}
 y_{1i} &| \eta_{1i} \sim \text{Po}(\lambda_i) \\
 \log(\lambda_i) &= \beta_{10} + f_{1d}(z_{di}) + f_{1y}(z_{yi}) + \alpha_{s1} \cdot u_i, \\
 y_{2j} &| \eta_{2j}, \tau \sim \text{Gamma}(y_i | \eta_{2j}, \tau), \\
 \log(\mu_j) &= \beta_{20} + f_{2d}(z_{dj}) + f_{2y}(z_{yj}) + u_j, \\
 y_{3j} &| \eta_{3j} \sim \text{Ber}(\pi_j), \\
 \text{logit}(\pi_j) &= \beta_{30} + \alpha_{d3} \cdot f_{2d}(z_{dj}) + f_{3y}(z_{yj}) + \alpha_{s3} \cdot u_j, \\
 y(s_j) &| \lambda(s_j) \sim \text{LGCP}(\lambda(s_j)), \\
 \log(\lambda(s_j)) &= \beta_{40} + \alpha_{d4} \cdot f_{2d}(z_{dj}) + u_j^*,
 \end{aligned}$$

where:

- the α components represent scaling parameters for the **shared effects**;
- f_d represents a **structured random effect related to the depth covariate** (a second-order random walk for f_{1d} and a one-dimensional SPDE for f_{2d});
- f_y is a **first-order random walk for years**; and
- u is a **separable type III spatio-temporal effect** (Knorr-Held, 2000), with a precision matrix derived from a **two-dimensional SPDE with Matérn covariances for the spatial part** and an **iid precision matrix for the temporal part**.

Results of the distribution of the hake in the bay of Biscay

- The first column in the left shows the **spatio-temporal effect** for the **first temporal node** of the spatio-temporal component for the **LGCP**



7 | Dealing with presence-only data



How to handle presence-only data

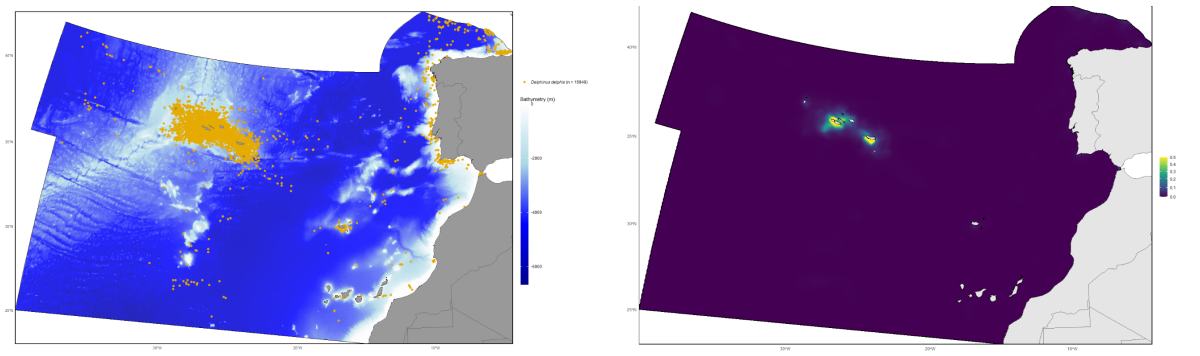
- In many situations our data are just a mere representation of places where we have seen the species of interest.
- We cannot say that it is not in the remaining places because we do not know whether the species is present or not. So there is no information about where the species is not.
- The usual approach is to generate some **pseudo-absences** (Fuster et al., 2024).
- This approach has many drawbacks: where should they be (although random, not clear), how many of them, in which proportion.
- Dorazio (2014) highlighted **various statistical models developed to integrate presence-only data** from different research protocols. But these models **often overlook the impact of imperfect detectability and survey bias**.
- A better choice is to **consider the observed locations as a point pattern**, that is, point events coming from Point process model (Warton and Shepherd, 2010; Renner et al., 2015).

How to handle presence-only data in fisheries context

- Yuan et al. (2017) introduced many useful tools for analyzing point patterns in INLA in the context of spatio-temporal distance sampling data from a large-scale survey of blue whales.
- After this work, Martino et al. (2021) demonstrated that **correcting for detectability issues** can mitigate biases in estimates caused by varying detection mechanisms during data collection, thereby **allowing for the integration of multiple information sources**.
- Pace et al. (2022) went further by providing a novel approach to **refine the presence-only detection function**, particularly using social media data sources, and by exploring potential seasonal effects, supported by a larger dataset and a broader study area.
- Still, the relatively **low number of sightings, dispersed over time, limits the ability to identify detailed temporal patterns**. Not space-time analysis.

Assessing cetacean spatial distribution from opportunistic sighting data

Fernández et al. (2025) have used this approach to model the **spatial distribution of common dolphin** (*Delphinus delphis*) and sperm whale (*Physeter macrocephalus*) using **Log-Gaussian Cox Process** with presence-only data and integrating a proxy for sampling effort.



Left: Study area with Dolphins sightseens in summer; right: prediction of the intensity (obs/km²).

8 | R shiny app



A Shiny app for implementing SDMs (Figueira et al., 2024b).

- **BAYSPINS** (BAYesian SPatial INla for SDMs) has been created to allow practitioners to perform:
 - ▶ geostatistical and preferential modeling;
 - ▶ a mixture of both modelings;
 - ▶ simulation of models to understand its behaviour;
 - ▶ **feedback analysis between models.**
- Can be used for **applied scientists INLA beginners**: the use of default settings automates the process or the customisation of a large number of elements that drive the modelling process.
- Allows from **quick initial evaluations to more rigorous studies depending on the user's skill** and understanding of the fundamentals underpinning the application.
- More info at <https://github.com/MarioFigueiraP/ShinyAppSpatialModelFeedback>.

9 | Conclusions



Conclusions

- Hierarchical Bayesian modelling can be a really useful tool for analysing SDMs.
- INLA + SPDE can also be very convenient as they are fast and provides good results (probably not the best ones, but the first ones in your analysis).
- Situations in real life bring us many other statistical issues that can be incorporated in the hierarchical framework presented: Non-stationarity, excess of zeroes, spatio-temporal models, misalignment and preferential sampling among others.
- Still many open problems! Extreme values, etc.

10 | References



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